BABU BANARASI DAS UNVERSITY LUCKNOW

SUBJECT NAME :- AEESCA

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Project - Section 1

Case Study : Protecting Healthcare data in Cloud.

Introduction - Healthcare organizations store sensitive patients data such as -

- Personal details (name, address, phone no.)
- Medical records (reports, diagnosis, prescriptions)
- Financial details.

Since this data is stored in the cloud, it becomes a target for cyber-attacks. Therefore, strong security techniques like anonymization, tokenization, DLP are required.

* **Data Protection Techniques** -

① **Data Anonymization** - It removes or modifies personal information so that a person can't be identified.

* **Techniques** -

- Masking → Hide data (eg. XXXX1234)
- Generalization → Age 25 → Age group (20 - 30)
- Suppression → Remove sensitive fields

Ex -

Original Data	Anonymized data
Rahul Sharma	Patient_01
Age : 25	Age 20 - 30

Advantages - ① Ensures privacy.

② Useful for research & analytics.

③ Reduces legal risks.

② **Tokenization** – It replaces sensitive data with a random token. The actual data is stored securely in a separate database.

Example –

Original data	Token
987001122	TKN56789

How it works?

User enter sensitive data
↓
System generates token
↓
Original data stored in secure server
↓
Token used for processing.

Advantages – ① Data is meaningless if stolen
② secure & reversible
③ Used in banking & healthcare.

③ **DLP (Data Loss Prevention)**: – DLP is a security system that detects & prevents data leakage.

Types of DLP –
- Network DLP – Monitors data over network.
- Endpoint DLP – Monitors devices (USB, Laptop)
- Cloud DLP – Protects cloud data.

Functions – ① Detect sensitive data
② Block unauthorized transfers
③ Send alerts to admin.

* Risks in Healthcare Cloud -

① Data breach - Hackers may steal patient records.

② Insider threat - Employees misuse data intentionally or accidentally.

③ Cloud Misconfiguration - Incorrect settings can expose data publicly.

④ Re-identification Risk - Even anonymized data can sometimes be traced back.

* Solutions / Security Measures -

• Technical solutions -

→ Encryption - Protect data during storage & transfer

→ Access control (RBAC) - limit user access.

→ MFA - extra login security

• Organizational measures -

→ Regular security audits

→ employee training

→ strong password policies.

• Cloud security -

→ Proper configuration

→ Continuous monitoring

→ Backup & recovery system

* Data Protection Flow

Patient data

↓

Anonymization

↓

Tokenization

↓

Cloud storage

↓

DLP

↓

Secure Data.

Conclusion -

Healthcare data is highly sensitive & must be protected.

Using anonymization, Tokenization & DLP together -

- Protect patient privacy.
- Prevents data leaks.
- Ensures secure data storage.

Section 2

Secure Cloud Architecture Design

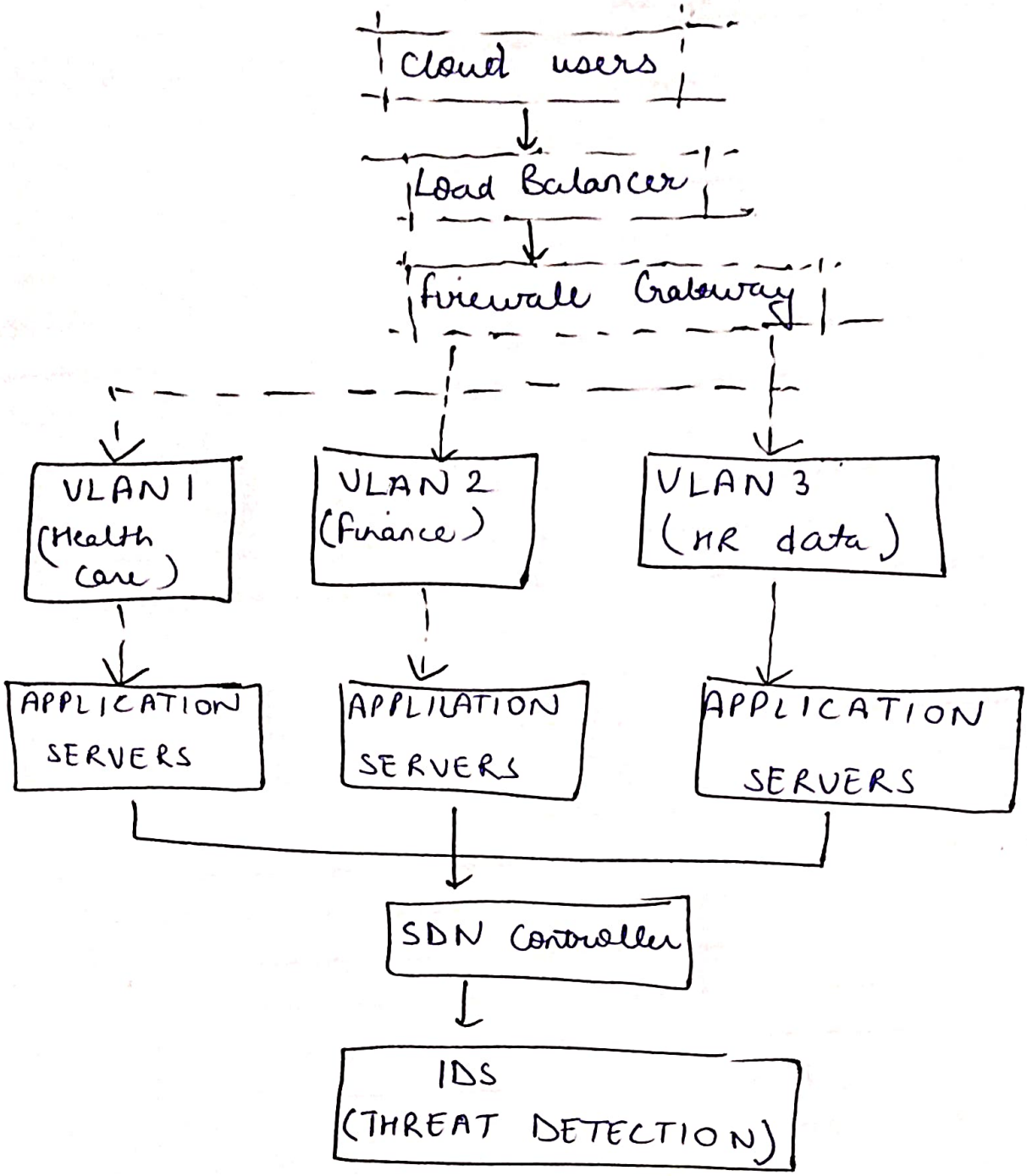
Introduction - Cloud-based organizations store & processes large amounts of data over the internet. This makes them vulnerable to cyberattacks such as - DDoS attacks, phishing, Malware, MitM.

To protect the system, a secure architecture using SDN, VLANs & IDS is required.

Core Components of Secure Architecture -

1. SDN (Software Defined Networking) -
2. VLAN (Virtual Local Area network)
3. IDS (Intrusion Detection System)
4. Firewall
5. VPN
6. Encryption
7. Load Balancer.

Secure Architecture Diagram -



Conclusion - Cloud security is critical in healthcare system. Data techniques like anonymization, DLP ensure privacy, while SDN, VLAN & IDS provide security.